

7BUIS009W Data Visualisation and Dashboarding

Module leader	Rolf Bänziger
Unit	Coursework
Weighting:	70%
Qualifying mark	40%
Description	Data Analysis, visualisation narrative and presentation
Learning Outcomes Covered in this Assignment:	This assignment contributes to obtaining the following LOs: • L01: discuss and critically apply the basic principles to data visualisation techniques; • L03: evaluate critically data visualisation using appropriate software tools; • L04: use a design process to build interactive Dashboards; • L05: synthesise the application of raw data into meaningful visualisation results and justify the appropriate techniques.
Handed Out:	5 th November 2024 (subject to minor changes after moderation)
Due Date	Date: 7th January 2025 Time: 13:00
Expected deliverables	Report Working files (separate link)
Method of Submission:	Online Submission via Blackboard Turnitin submission as a PDF document in the format: 7BUIS009W_StudentNumber_firstName_lastName.pdf
Type of Feedback and Due Date:	Feedback after submission Written feedback on the CW will be provided within three weeks after the submission (unless unexpected circumstances occur). All marks will remain provisional until formally agreed by an Assessment Board.

Assessment regulations

See the Assessment guidelines <https://www.westminster.ac.uk/current-students/guides-and-policies/assessment-guidelines> for a clarification of how you are assessed, penalties and late submissions, **what constitutes plagiarism etc.**

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in the range 50-59%, in which case the mark will be capped at the pass mark (50%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognised that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University Assessment Regulations, please refer to the following website:

<http://www.westminster.ac.uk/study/current-students/resources/academic-regulations>

Note: By submitting the work through Blackboard you are acknowledging that this is solely your own work. Any code which is not created by you MUST be clearly commented as such. Any code discovered to not have been created by you will mean that the work will be submitted to academic standards for a potential assessment offence, which may result in a zero mark in the component or whole module.

Coursework Description

Using one or more publicly available data sets (e.g. from data.gov.uk) to:

1. Formulate a research question – this will frame your analysis and is the title of your report.
2. Carry out exploratory data analysis of the data.
3. Using the findings from your analysis, construct a data story to inform the viewer of the results of your data analysis and how you have interpreted them.
4. Write a critical reflection.

Report

You need to submit your work as a report. Your report need to contain two parts:

Part A is the data story, which presents your findings. You should use storytelling techniques to tell and visualise your findings. It needs to be 800 words (+/- 10%) and contain 6 visualisations. A visualisation can be either a single chart, an infographic or a dashboard.

Part B is a critical reflection of your work. You need to state:

- What data did you use? Why and how did you select your sources?
- How did you prepare/transform/check your data?
- What did you do well?
- What could you improve in your story and visualisations?

Part B must not exceed 800 words.

Supporting files

In addition to the report, you must also submit all the files you created during your project. That includes any manipulated data files, Power BI and R files, and exported graphics. You do not need to submit the original data sets, but you must clearly indicate where they are available. Submissions without complete supporting files might not be marked.

Frequently asked questions

Can I use a spreadsheet tool like Excel?

Yes, but ONLY for data preparation and cleaning. You may NOT use it for the exploratory data analysis or producing the final data graphics.

Can I use other tools than Tableau, Power BI or R to create graphics?

Yes, you can use any other professional tool to create graphics, but not Excel. If you're in doubt what consists a "professional data visualisation tool", consult with your module leader.

Can I have the same research questions as another student?

Each student must come up with an individual research question. While two or more students could have conceived the same research questions, this is very rare, and we would take great care to affirm that each student submitted original work.

Indicative Marking criteria

Data story narrative (30 points)

1. Research Question Relevance and Clarity (5 marks):

- The data story effectively addresses the initial research question, providing a clear and comprehensive understanding of its significance.

2. Coherence and Flow of the Narrative (10 marks):

- The narrative demonstrates a coherent flow, connecting each section seamlessly and facilitating a smooth progression of ideas.
- The data story engages the viewer through compelling storytelling techniques, leaving a significant impact and fostering a deeper understanding of the subject matter.

3. Data Interpretation and Insights (10 marks):

- The data story offers insightful interpretations of the findings, presenting meaningful insights derived from the exploratory data analysis.

4. References (10 marks)

- References are used where necessary (in particular, but not exclusively for the used data)
- In-text citations and bibliography are formatted correctly.

Visualisations (40 points)

1. Data Representation (12 points):

- The visualisations accurately represent the data without distorting or misinterpreting it.
- The chosen visualisations are appropriate for the type of data being presented.
- Visualisations effectively convey the key insights and trends present in the dataset.

2. Clarity and Interpretability (8 points):

- Visualisations are clear and easy to understand, even for individuals not familiar with the dataset.
- Labels, legends, and annotations are used effectively to enhance the interpretability of the visualisations.
- Complex information is presented in a simplified and easily digestible manner.

3. Aesthetics and Design (6 points):

- Visualisations exhibit a visually appealing and professional design.
- Colour choices are appropriate and enhance the overall readability of the visualisations.
- The layout and formatting of the visualisations are visually engaging and contribute to the overall story.

4. Relevance to Data Story (6 points):

- The visualisations directly support the data storytelling and contribute to a comprehensive understanding of the topic.
- Each visualisation provides meaningful insights that contribute to the narrative of the data story.

5. Innovation and Creativity (6 points):

- The visualisations demonstrate creative approaches to presenting the data, showcasing originality and innovative design choices.
- Unique visualisation techniques are employed to convey complex information in a compelling manner.

6. Accuracy and Completeness (4 points):

- The visualisations accurately represent all relevant aspects of the data analysis without any major errors or omissions.
- The visualisations provide a comprehensive view of the data story, leaving no significant gaps in the analysis.

7. Consistency and Cohesion (4 points):

- Visualisations are consistent in style and formatting, maintaining a cohesive look throughout the data story.
- The visualisations complement each other and contribute to a unified and coherent narrative.

Critical Reflection (30 points)

1. Clarity of Data Selection (10 marks):

- The critical reflection clearly explains the rationale behind the choice of data sources and justifies their relevance to the research question.

2. Thoroughness of Data Preparation (10 marks):

- The critical reflection demonstrates a comprehensive understanding of the steps taken to prepare, transform, and check the data for analysis.

3. Identification of Strengths and suggestions for Improvements (10 marks):

- The critical reflection effectively identifies and elaborates on the aspects of the analysis that were executed well, showcasing a nuanced understanding of successful strategies.
- The critical reflection provides insightful suggestions for enhancing the data story and visualizations, demonstrating a critical approach to self-improvement.

Word count (marks will be deducted)

Word count: No more than 10% above or beyond the prescribed word count, otherwise 10 marks will be deducted for each 10% under or over the requirement.

Word count in Part B: No more than 10% above the prescribed word count, otherwise 10 marks will be deducted for each 10% over the requirement (no minimum word count).